TABLE S2. Experimental details and results including target packages materials (PC and SC refer to stishovite polycrystal and single crystal, hdC for single crystal diamond) and thicknesses in μm, drive energy, measured shock velocities in the quartz and the stishovite and the inferred pressure and density from the impedance match analysis. VISAR sensitivities (vacuum velocity per fringe:vvpf) were 16.083 and 6.907km/s for shots 68849-70427 and 4.104 and 9.951km/s for shot 70912 and 70917. The refractive index used to obtain the true shock velocity from the apparent shock velocity was $n = 1.799 \pm 0.002$ for the single crystal and $n = 1.82 \pm 0.02$ for the polycrystals (see S8)

ShotID	Package (μm)	Energy (J)	$U_{S\ Qtz}$ (km/s)	U_S (km/s)	u_P (km/s)	P (GPa)	ρ (g/cm^3)
68849	Be[26.5]Au[9.5]Qtz[99]PC[301]	3715	27.23 ± 0.32	26.6 ± 0.39	14.93 ± 0.33	1702 ± 38.4	9.78 ± 0.37
68850	Be[26.5]Au[9.5]Qtz[103]PC[299]	1921	24.64 ± 0.37	24.01 ± 0.34	13.14 ± 0.36	1352.5 ± 37.6	9.48 ± 0.38
68851	Be[26.5]Au[9.5]Qtz[97.5]PC[299]	5920	30.81 ± 0.35	29.80 ± 0.45	17.48 ± 0.37	2232.5 ± 48.9	10.38 ± 0.43
68852	Be[26.5]Au[9.5]Qtz[35.5]PC[299]	982	23.01 ± 0.39	23.07 ± 0.38	11.87 ± 0.37	1174.0 ± 37.6	8.84 ± 0.36
68854	Be[49]Au[9.5]Qtz[35.5]SC[93]Qtz[114]	1967	32.70 ± 0.36	31.25 ± 0.43	18.86 ± 0.37	2526.7 ± 51.3	10.82 ± 0.45
70426	hdC[20]Au[3]Qtz[90.5]SC[300]	952	21.82 ± 0.24	21.65 ± 0.20	11.12 ± 0.21	1032.2 ± 20.2	8.82 ± 0.22
70427	hdC[19.5]Au[3]Qtz[35]SC[247]	1300	29.38 ± 0.27	28.72 ± 0.21	16.39 ± 0.27	2017.4 ± 34	9.99 ± 0.26
70912	hdC[23.5]Au[3]Qtz[37.5]SC[314]	1717	32.70 ± 0.22	32.15 ± 0.15	18.58 ± 0.23	2561.4 ± 31.4	10.16 ± 0.19
70917	hdC[22]Au[3]Qtz[38]SC[509]	1832	33.31 ± 0.38	32.73 ± 0.23	18.97 ± 0.36	2660.4 ± 47.8	10.21 ± 0.30